NAME: ____

PERIOD: ________ DATE:_______________________

SEISMIC SLINKY: P & S SEISMIC WAVES

www.iris.edu/earthquake

Analysis: Answer the following questions using complete sentences.

1. Draw what you see the waves doing:

P (Compressional)	S (Transverse)

2. Which one is faster? Why?

3. Which wave has the greatest amplitude?

4. Does the Slinky© model solids, liquids, or both?

5. What are seismic waves?

6. Draw a picture of a simple waveform in the box:

7. List four wave characteristics below and write a short description in your own words. Once you have done this, label the parts of the wave on your drawing of a transverse wave (not compressional!)

1. ________________________ — ____

2. ________________________ — ____

3. ________________________ — ____

4. ________________________ — ____

INSTRUCTOR ANSWER KEY

NAME: ______________________________

PERIOD: ________ DATE: ______________________

www.iris.edu/earthquake

SEISMIC SLINKY: P & S SEISMIC WAVES

Analysis: Answer the following questions using complete sentences.

1. Draw what you see the waves doing:

P (Compressional):	S (Transverse):

2. Which one is faster? Why?

- P-waves travel 60% faster than S-waves on average. Compression waves apply a force in the direction of travel which is faster than the perpendicular motion of the S-wave. The S wave may appear faster in the demo because there is less friction in the perpendicular motion of the Slinky© coils. The P wave has more drag on the floor.

3. Which wave has the greatest amplitude?

- Between P and S waves, the S-waves have the greatest amplitude (Surface waves have greater amplitude than S waves.) S waves are shear waves, which means that the motion of the medium is perpendicular to the direction of travel of the wave. The energy is thus less easily transmitted through the medium, and S-waves are slow and higher.

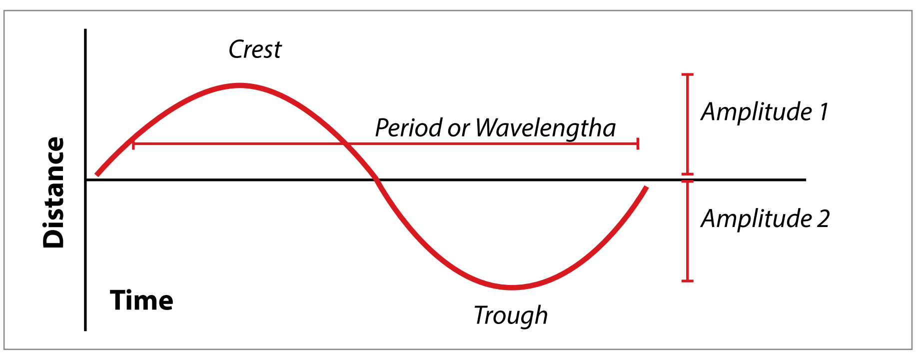
4. Does the Slinky© model solids, liquids, or both?

- The Slinky© models only seismic waves through solids. Why? Because the molecules are coupled together. To model seismic waves through liquids, the "Human Wave" is a good analogy.

5. What are seismic waves? possible answers:

- An elastic wave in the earth produced by an earthquake or underground explosion.
- A wave of energy that is generated by an earthquake or other earth vibration.
- Any of several forms of vibrational waves that travel through the Earth as the result of an earthquake.

6. Draw a picture of a wave in the box



7. List the wave characteristics below and write a description in your own words. Once you have done this, label the parts of the wave on your drawing of a transverse wave (not compressional!)

1. Period (or Wavelength) — distance from crest to crest, trough to trough, **OR** from a pt on one wave cycle to the corresponding pt on the adjacent wave cycle. **OR**... time it takes a particle to complete one full cycle

2. Amplitude — The height between the mean (zero) and the crest (high) or the trough (low).

3. Crest — The highest point on any wave

4. Trough — The lowest point on any wave